Mathematics A

Unit 2 • Chapter OL-T23 • Expertise Q16+ Classified

Cross-session • chapter-organised candidate

5 classified items • 16 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Expertise • Unit 2 • Chapter OL-T23 • Expertise Q16+ • 5 items • 16 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebraic Proof

Unit 2 • Expertise • 16 marks • 5 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

4MA1-2H Q18

3 marks

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

WORKING SPACE

4MA1-2H Q18 **3 marks**

18 Prove that when the sum of the squares of any two consecutive odd numbers is divided by 8, the remainder is always 2
Show clear algebraic working.

Chapter: Algebraic Proof **3 marks**

Q18 – Chapter: Algebraic Proof

Family: Prove divisibility or a remainder • Variant: Fixed remainder modulo a divisor

Method
Let the consecutive odd numbers be $2n-1$ and $2n+1$.
Why: Any two consecutive odd numbers have this form.

Method
Their square sum is (2n−1)^2+(2n+1)^2=8n^2+2.
Why: Expand and add.

Method
$8n^2$ is divisible by 8, so the remainder is always 2.
Why: Separate the multiple of 8 from the remainder.

Final answer
remainder: 2

January 2020 | 4MA1-2H | Question 17 | 3 marks

Chapter OL-T23: Algebraic Proof

17 Prove that the difference between two consecutive square numbers is always an odd number.
Show clear algebraic working.

Worked solution

January 2020 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Consecutive integer parity identity

Method / working

1. Let consecutive squares be n^2 and $(n+1)^2$.
2. Difference $=(n+1)^2-n^2=2n+1$.
3. $2n+1$ is always odd for integer n.

Final answer: $(n+1)^2-n^2=2n+1$, which is odd

June 2021 | 4MA1-1H | Question 17 | 3 marks

Chapter OL-T23: Algebraic Proof

17 Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

Worked solution

June 2021 | Question 17 | 3 marks

Family: Prove a universal parity statement • Variant: Consecutive even/odd sequence proof

Method / working

1. Represent consecutive even numbers as 2n, 2n+2, 2n+4.
2. Subtract squares: $(2n+4)^2-(2n)^2 = 16n+16$.
3. Factor: $16n+16 = 8(2n+2)$, which proves the result.

Final answer: Proof: difference = $8(2n+2)$

19 *ABCED* is a five-sided shape.

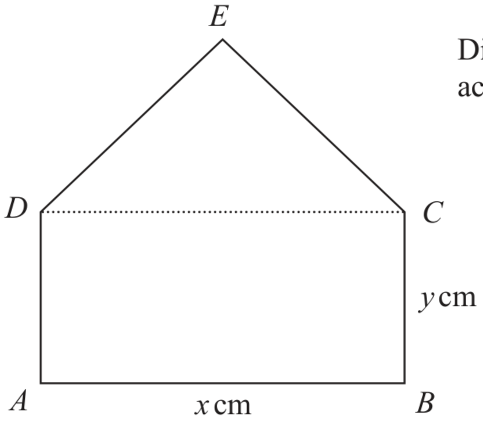

Diagram **NOT** accurately drawn

ABCD is a rectangle.
CED is an equilateral triangle.

$AB = x$ cm $\quad BC = y$ cm

The perimeter of *ABCED* is 100 cm.
The area of *ABCED* is R cm^2

(a) Show that $R = \frac{x}{4}\left(200 - \left[6 - \sqrt{3}\right]x\right)$

(3)

(b) (i) Find the value of x for which R has its maximum value.

Give your answer in the form $\frac{p}{q - \sqrt{3}}$ where p and q are integers.

$x =$ ....

(2)

(ii) Explain why the maximum value of R is given by this value of x.

....

....

....

(1)

Worked solution

November 2021 | Question 19 | 3 marks

Family: Construct an algebraic identity proof • Variant: Derive a general geometric algebra formula

Method / working
1. The perimeter is made from three sides of length \(x\) and two sides of length \(y\):
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
2. 3x+2y=100
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
3. So
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
4. y=frac{100-3x}{2}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
5. The area is the area of the rectangle plus the area of the equilateral triangle:
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
6. R=xy+frac{\sqrt3}{4}x^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
7. Substitute \(y=\dfrac{100-3x}{2}\):
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
8. R=x\left(frac{100-3x}{2}\right)+frac{\sqrt3}{4}x^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
9. R=50x-frac32x^2+frac{\sqrt3}{4}x^2
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
10. R=frac{x}{4}\left(200-(6-\sqrt3)x\right)
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
11. For the maximum, write
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
12. R=-frac{6-\sqrt3}{4}x^2+50x
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
13. The maximum occurs at
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
14. x=frac{-b}{2a}=frac{-50}{2\left(-frac{6-\sqrt3}{4}\right)}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
15. x=frac{100}{6-\sqrt3}
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
16. The coefficient of \(x^2\) is negative, so the quadratic graph opens downwards and this vertex gives a maximum.
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
17. Answer: \(R=\dfrac{x}{4}\left(200-(6-\sqrt3)x\right)\), and \(x=\dfrac{100}{6-\sqrt3}\).
Reason: Apply the stated definition, theorem, algebraic operation, or diagram convention.
Final answer: R=\dfrac{x}{4}\left(200-(6-\sqrt3)x\right), and x=\dfrac{100}{6-\sqrt3}.

17 Given that $8\sqrt{m} + \sqrt{49m} - \sqrt{9m} = k\sqrt{m}$

where k is an integer and m is a prime number,

(a) work out the value of k

$k =$ ....

(1)

(b) Show that $\dfrac{5 - \sqrt{18}}{1 - \sqrt{2}}$ can be written in the form $a + b\sqrt{2}$

where a and b are integers.
Show each stage of your working clearly.

(3)

(Total for Question 17 is 4 marks)

Answer reference | January 2023 | Question 17

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Question		Working	Answer	Mark		Notes
17	(a)		12	1	B1	
	(b)	$\frac{5-\sqrt{18}}{1-\sqrt{2}}\times\frac{1+\sqrt{2}}{1+\sqrt{2}}$ **or** $\frac{5-\sqrt{18}}{1-\sqrt{2}}\times\frac{-1-\sqrt{2}}{-1-\sqrt{2}}$ oe		3	M1	Multiplying numerator and denominator by $1+\sqrt{2}$
		$\frac{5-\sqrt{36}+5\sqrt{2}-\sqrt{18}}{1+\sqrt{2}-\sqrt{2}-2}$ **or** $\frac{5-6-3\sqrt{2}+5\sqrt{2}}{-1}$ **or** $\frac{-5+6+3\sqrt{2}-5\sqrt{2}}{1}$ oe NB:allow $\sqrt{18}$ or $3\sqrt{2}$ $\sqrt{36}$ or 6 or $\sqrt{6}\sqrt{6}$			M1	Showing correct expansions (not necessarily as a fraction)
		working required	$1-2\sqrt{2}$		A1	dep on M2 (ie all stages of working must be shown convincingly) or for stating $a = 1$ and $b = -2$
						Total 4 marks

Worked solution

January 2023 | Question 17 | 4 marks

Family: Prove a universal parity statement • Variant: Product or sum of arbitrary odd/even numbers

Official final mark-scheme pages are embedded immediately after this question.